Mahmoud Alyosify

AI Agent Systems Engineer

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SUMMARY

Engineer who builds **AI agents that finish multi-step work**: tool use, routing, self-checks and shared memory. Currently building a LangGraph RAG agent behind a FastAPI service at Injaz Tech; designed a multi-agent intelligence platform over a knowledge graph. MSc in Artificial Intelligence, Queen's University, Canada (GPA 4.30 / 4.30).

SKILLS

Agents: LangGraph, LangChain, tool and function calling, structured outputs, multi-agent systems, RAG
LLMs: OpenAI and Anthropic APIs, Ollama, prompt engineering, fine-tuning (LoRA)
Engineering: Python, FastAPI, Pydantic, pytest, Docker, AWS
Data: FAISS, ChromaDB, MongoDB, SQL

EXPERIENCE

AI Engineer Intern — Injaz Tech (Injaz Group), Cairo — Oct 2026 – Present

- Built a **LangGraph** agent that routes each question, grades the evidence it finds, rewrites weak queries and checks its own answer.
- Raised top-1 retrieval accuracy from **75% to 94%** and cut false refusals from 5 in 20 to 1 in 20.
- Built the **FastAPI** service: typed request and response models (Pydantic), streaming answers, session memory, health checks and rate limiting.

Part-Time Machine Learning Instructor — National Telecommunication Institute (NTI), Egypt — Jul 2025 – Present

- Teach **90-hour machine learning bootcamps** and AI for Business modules in NTI's Summer Internships Program.

Systems Engineer (Military Service) — National Company for Roads Building & Development, Cairo
Feb 2024 – Mar 2025

- Supported the backend systems of Egypt's national toll network, used by **more than half of the country's vehicles**.

Online Instructor — Udemy & YouTube (Arabic courses), Remote — 2020 – Present

- Created Arabic courses on algorithms, discrete mathematics, probability and statistics: **10,000+ learners, 700K+ views**.

PROJECTS

HORUS Sentinel — multi-agent intelligence platform (Python, LLM agents, knowledge graph)

- Designed a team of AI agents that keep collecting open-source information into a **knowledge graph**; a self-hosted LLM writes evidence-backed reports from it.
- Each agent owns one source; every finding is linked to earlier evidence in the shared graph, so reports trace each claim back to its source.

Automated Multimodal Agent — PDF to narrated slides (Python, LLM APIs, text-to-speech)

- An agent that turns a PDF into a narrated PowerPoint deck: it reads the document, plans and writes the slides, and records the narration.

Human vs. AI-Agent Pull Requests — research study (Python, GitHub data mining)

- Built a reproducible pipeline that mines GitHub to compare review effort on pull requests written by AI agents and by people.

EDUCATION

MSc in Artificial Intelligence — Queen's University, Kingston, Canada — Sep 2025 – Oct 2026

- GPA **4.30 / 4.30** · Completed October 2026 · Digilians Presidential Scholarship

BSc in Computer & Information Science (Bioinformatics) — Assiut University, Egypt — Sep 2019 – Jul 2023

- GPA 3.53 / 4.00 (Very Good with Honours) · Graduation project graded A+

CERTIFICATIONS & AWARDS

IBM AI Developer Professional Certificate (IBM, 2026) · Deep Learning Specialization (DeepLearning.AI, 2026) · 1st Place, Smart Cities Hackathon 2022 (77 university teams)

LANGUAGES

Arabic (native) · English (professional working proficiency)